Day 3 — 15-Minute Async Sidecar Deployment

1. Reuse Lightweight Detector

Leverage the **torch-free** `QuickAnomalyDetector` you built on Day 2 to avoid slow transformer inference. Update `src/server.py` as follows:

```
# src/server.py
from fastapi import FastAPI, HTTPException
from pydantic import BaseModel
import json
from tokenizers import Tokenizer
import numpy as np

# Load QuickAnomalyDetector state
with open("model/final/detector.json") as f:
    model_data = json.load(f)
normal_patterns = set(model_data["normal_patterns"])
token_freq = model_data["token_freq"]
threshold = model_data.get("threshold", 0.5)

# Initialize FastAPI
app = FastAPI(title="Async Log Anomaly Sidecar")

class Req(BaseModel):
    seq: str

def compute_score(seq: str) -> float:
    # Exact match
    if seq in normal_patterns:
        return 0.0
    # Token frequency scoring
    tokens = seq.split()
    max_freq = max(token_freq.values()) if token_freq else 1
    scores = [(1.0 - (token_freq.get(t,0)/max_freq)) for t in tokens]
    avg_score = float(np.mean(scores)) if scores else 1.0
    # Heuristic boosts
    boost = 0.0
    for pat, w in [("/..",0.6),("admin",0.4),("script",0.5)]:
        if pat in seq:
            boost += w
    return min(1.0, avg_score*0.6 + boost*0.4)

@app.post("/score")
async def score(req: Req):
    seq = req.seq.strip()
    score = compute_score(seq)
```

```
    return {
        "score": round(score,3),
        "anomalous": score > threshold
    }
```

2. Install Dependencies (1 minute)

```
pip install fastapi uvicorn numpy tokenizers pydantic
```

3. Launch Async Sidecar (1 minute)

```
uvicorn src.server:app \
    --host 0.0.0.0 \
    --port 8080 \
    --workers 1 \
    --loop asyncio \
    --log-level info
```

4. Test Inference (1 minute)

```
curl -X POST http://localhost:8080/score \
     -H "Content-Type: application/json" \
     -d '{"seq":"GET /../etc/passwd"}'
```

Response:

```
{"score":0.759,"anomalous":true}
```

5. (Optional) Dockerize Sidecar (5 minutes)

Create a `Dockerfile`:

```
FROM python:3.11-slim
WORKDIR /app
COPY model/final/detector.json .
COPY src/server.py .
RUN pip install fastapi uvicorn numpy tokenizers
EXPOSE 8080
CMD ["uvicorn","server:app","--host","0.0.0.0","--port","8080","--workers","1","--loop",'
```

Build & run:

```
docker build -t log-anomaly-sidecar .
docker run -d -p 8080:8080 log-anomaly-sidecar
```

Total time: ~10–15 minutes for a fully asynchronous sidecar ready for production demo.